

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record
Practice Program

Date: 17/07/2026

Week No.: 2

Program S. No. 3

Title	Student Marks Data Manipulation and Analysis using Pandas																																	
Problem Statement	Data Cleaning and Preprocessing with Pandas: Create a DataFrame with some missing values in columns Name, Age, and City. Perform the following operations: fill missing values in the Age column with the mean age, drop rows where Name or City is missing, and convert the Age column to integer type. Finally, normalize the Age column using min-max scaling.																																	
Objectives	- To understand how to create a Pandas DataFrame containing missing values. - To learn how to identify and handle missing data using the fillna() function. - To remove incomplete records using the dropna() function. - To convert a DataFrame column to the required data type using the astype() function. - To normalize numerical data using the Min-Max Scaling technique. To understand the importance of data preprocessing before performing data analysis and machine learning tasks.																																	
Dataset URL Link / File	<table><tr><th>Name</th><th>Age</th><th>City</th></tr><tr><td>Arjun</td><td>np.nan</td><td>np.nan</td></tr><tr><td>Priya</td><td>28</td><td>Mumbai</td></tr><tr><td>Rohan</td><td>35</td><td>Delhi</td></tr><tr><td>Sneha</td><td>np.nan</td><td>np.nan</td></tr><tr><td>Amit</td><td>40</td><td>Bangalore</td></tr><tr><td>Deepa</td><td>30</td><td>Chennai</td></tr><tr><td>Rahul</td><td>np.nan</td><td>np.nan</td></tr><tr><td>np.nan</td><td>33</td><td>Kolkata</td></tr><tr><td>np.nan</td><td>29</td><td>Ahmedabad</td></tr></table>				Name	Age	City	Arjun	np.nan	np.nan	Priya	28	Mumbai	Rohan	35	Delhi	Sneha	np.nan	np.nan	Amit	40	Bangalore	Deepa	30	Chennai	Rahul	np.nan	np.nan	np.nan	33	Kolkata	np.nan	29	Ahmedabad
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np.nan	29	Ahmedabad																																
Functions Details	Sl. No.	Function	Package	Description of function																														
	1	DataFrame()	pandas	Creates a DataFrame from structured data																														
	2	Mean()	pandas	Calculates the average value																														
	3	fillna()	pandas	Replaces missing values with a specified value.																														
	4	dropna()	pandas	Removes rows containing missing values.																														

Student Reg. No: 26MML0031 Name: Makada Hasnain Husainbhai

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Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1 Import the pandas library. Step 2 Create a Python dictionary containing StudentID, Name, and Marks for five students. Step 3 Convert the dictionary into a Pandas DataFrame and display the original dataset. Step 4 Select and display only the Name and Marks columns. Step 5 Filter and display students whose Marks are greater than 70. Step 6 Add a Grade column by assigning grades A to E according to the marks obtained. Step 7 Display the updated DataFrame containing the Grade column. Step 8 Sort the DataFrame by Marks in ascending order and display the result. Step 9 Calculate and display the average marks using the mean() function.
Program	<pre># Import the Pandas library import pandas as pd # Create a dictionary containing student details dictionary_1 = { "StudentID": [10, 11, 12, 13, 14], "Name": ["Hasnain", "John", "Alice", "Sahil", "Cook"], "Marks": [45, 86, 90, 67, 97] } # Convert the dictionary into a Pandas DataFrame dataFrame = pd.DataFrame(dictionary_1) # Display the original dataset print("Displaying the Original Dataset\n") print(dataFrame, "\n") # Select and display only the Name and Marks columns only_name_colum = pd.DataFrame(dataFrame, columns=["Name", "Marks"]) print("Displaying only name and marks columns\n") print(only_name_colum, "\n") # Filter students whose marks are greater than 70 marks_greater_than_70 = pd.DataFrame(dataFrame[dataFrame["Marks"] > 70]) print("Displaying students with marks greater than 70\n") print(marks_greater_than_70, "\n") # Assign grades based on the marks obtained dataFrame["Grade"] = ["A" if marks >= 90 else "B" if marks >= 80 else "C" if marks >= 70 else "D" if marks >= 60 else "E" for marks in dataFrame["Marks"]]</pre>

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```
]

# Display the updated dataset with the Grade column
print("Displaying the updated dataset\n")
print(dataFrame, "\n")

# Sort the dataset by Marks in ascending order
sort_by_marks = dataFrame.sort_values(
    by="Marks", ascending=True
)
print("Displaying the sorted dataset\n")
print(sort_by_marks, "\n")

# Calculate and display the average marks
average = dataFrame["Marks"].mean()
print("Average Marks \n")
print(average)
```

Displaying the Original Dataset

```
***
   StudentID  Name  Marks
0         10  Hasnain    45
1         11   John    86
2         12   Alice    90
3         13   Sahil    67
4         14   Cook    97
```

Figure 2.1 Original Student Dataset

2.2 shows only the Name and Marks columns selected from the original DataFrame. This demonstrates column selection and allows the analysis to focus on the student names and their corresponding marks without displaying StudentID.

Displaying only name and marks columns

```
   Name  Marks
0  Hasnain    45
1   John    86
2   Alice    90
3   Sahil    67
4   Cook    97
```

Figure 2.2 Selected Name and Marks Columns

Displaying students with marks greater than 70

```
   StudentID  Name  Marks
1         11   John    86
2         12   Alice    90
4         14   Cook    97
```

Figure 2.3 Filtering student with marks > 70

2.4 displays the updated DataFrame after the Grade column. Grades are assigned conditional logic: A for marks of 90 or B for 80-89, C for 70-79, D for 60-69, and marks below 60.

Displaying the updated dataset

```
   StudentID  Name  Marks Grade
0         10  Hasnain    45    E
1         11   John    86    B
2         12   Alice    90    A
3         13   Sahil    67    D
4         14   Cook    97    A
```

Figure 2.4 Dataset after Adding the Grade Column

Displaying the sorted dataset

```
   StudentID  Name  Marks Grade
0         10  Hasnain    45    E
3         13   Sahil    67    D
1         11   John    86    B
2         12   Alice    90    A
4         14   Cook    97    A
```

Figure 2.5 Dataset Sorted by Marks in Ascending Order

Figure 2.1 displays the original DataFrame created from a student dictionary. It contains the columns StudentID, Name, and Marks for five students. The pd.DataFrame function converts the dictionary into a structured format that can be analysed using Pandas.

Figure 2.3 presents the filtered records for students whose Marks are greater than 70. The Boolean condition `dataFrame["Marks"] > 70` retains John, Alice, and Cook because their marks satisfy the specified condition.

Figure 2.5 shows the records sorted from the lowest to the highest marks using `sort_values()`. Hasnain appears first with 45 marks, while Cook appears last with 97 marks. The original row indexes are retained by Pandas.

**Output
Snapshots
With
Explanation/
Observations**

Student Reg. No: 26MML0031 Name: Makada Hasnain Husainbhai

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Figure using five of	2.6 displays the average marks calculated using the mean() function. The total marks of all students divided by five gives an average of 77.0. <i>Figure 2.6 Average Marks</i>
Conclusions	This exercise successfully demonstrated basic student-data manipulation and analysis using Pandas. A DataFrame containing StudentID, Name, and Marks was created from a Python dictionary, and multiple operations were performed on the dataset. The results showed that: • The Name and Marks columns were selected successfully for focused analysis. • Students with marks greater than 70 were identified using Boolean filtering. • A Grade column was generated using conditional logic based on each student's marks. • The records were arranged in ascending order of marks using sort_values(). • The average marks of the five students were calculated as 77.0 using mean(). Overall, the exercise provided practical experience in DataFrame creation, column selection, row filtering, conditional transformation, sorting, and statistical aggregation. These operations are fundamental for data preprocessing and exploratory data analysis in Data Science and Machine Learning applications.
References	1. Wes McKinney, Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter, 3rd Edition, O'Reilly Media, 2022. 2. Pandas Documentation: https://pandas.pydata.org/docs/ 3. Python Documentation: https://docs.python.org/3/ 4. Python Program (Google Colab Link) https://colab.research.google.com/drive/1qBzvtH6ZEis8Jv20WDiYXKI4PXGSXcWJ?usp=sharing 4. Pandas DataFrame API Reference: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html 5. Class lecture link: https://drive.google.com/file/d/15diEukCtrPLimox63oa9ce1U-t4qVMUy/view

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